Chapter 3 Note

3.1

In the proof of Theorem 3.1, here is why the class of all transitive natural numbers is a set. We can show that it is a subset of ω, and is hence a set by A_2. For suppose x is a member of the class of all transitive natural numbers. Then obviously x is also a natural number, hence $x \in \omega$, thus completing the proof.

Note that here is why the only natural numbers allowed for by Definition 2.2 are the ones defined right before Definition 1.1. By Definition 2.2, a natural number is a member of every inductive set. Now, consider the following inductive set: $\emptyset$ is an element, and if x is an element, then so is x^+. Hence every natural number is a member of this set. But this set is just the set of all sets that were defined in the way just before Definition 1.1.

Exercise 3.1

Suppose that A is a class of natural numbers such that for every natural number n, if A contains all numbers less than n, then A contains n as well. Now, we can prove by induction that A contains all natural numbers. Consider the property P meaning A contains all numbers natural numbers less than or equal to n. Now, there are no natural numbers less than 0. Thus the statement that A contains all natural numbers less than 0 is vacuously true; hence A contains 0. Thus, because 0 is the only natural number less than or equal to 0, P holds for 0. Now suppose that P holds for some n, that is, A contains all natural numbers less than or equal to n. But then A contains all numbers less than $n+1$, meaning (by the definition of A) A also contains $n+1$. Thus, by induction, A contains all natural numbers.

Exercise 3.2

Consider a natural number n.

(a): Suppose that $x \in n$. Now, it cannot be that $n = \emptyset$. Thus n is non-empty, hence of the form $(n-1)^+$. Thus either $x \in n-1$ or $x = n-1$. Now, $\cup n$ is just the set of all elements of all elements of $n-1$ together with all elements of $n-1$. Now suppose $x = n-1$. Then $x \notin \cup n$. For suppose it were. Then it would either be an element of $n-1$ or it would be an element of an element of $n-1$. In the first case, we would have $n-1 \in n-1$, which is ruled out by Theorem 3.2. In the second case, there would be some y such that $x \in y$ and $y \in n-1$. But then, by the transitivity of $n-1$ (by Theorem 3.1), $x \in n-1$, that is, $n-1 \in n-1$, which is again ruled out by Theorem 3.2. Thus $n-1 \notin \cup n$, meaning the statement is not necessarily true.

(b): Suppose that $x \in \cup n$. Thus either $x \in n-1$ or x is an element of some element of $n-1$. In the

first case, because $n-1 \in n$, $x \in n$ (by the transitivity of n). In the second case, because $n-1$ is transitive, it follows that $x \in n-1$. But then, because $n-1 \in n$ and n is transitive, $x \in n$. Thus, in either case, $x \in n$, and thus $\cup n \subseteq n$. Hence the statement is necessarily true.

(c): By part (b) of this problem, it follows that $x \in \cup n$ implies $x \in n$. But $n^+ = n \cup \{n\}$. Thus all elements of n are in n^+, meaning $x \in n^+$. On the other hand, suppose that $x \in n^+$. Then it could be that $x = n$. But then, if $x \in \cup n$, we would have either n being an element of an element of n, or $n \in n$. In the first case, the transitivity of n would show that $n \in n$. Thus we would have, in either case, $n \in n$, which is impossible. Hence it is not necessarily the case that $n^+ \subseteq \cup n$. Thus $\cup n = n^+$ does not necessarily hold.

(d): On the one hand, suppose that $x \in \cup n^+$. Then either x is an element of an element of n, or x is an element of n. In the first case, because n is transitive, $x \in n$. The second case means $x \in n$. Thus, in any case, $x \in n$. On the other hand, suppose that $x \in n$. Then $x \in \cup n^+$, as $n^+ = n \cup \{n\}$. Hence $\cup n^+ = n$.

(e): By part (b) of this problem, $\cup n \subseteq n$ holds. But by the answer to part (a) of this problem, it does not necessarily follow that $n \subseteq \cup n$. Thus it is not necessarily true that $\cup n = n$.

Exercise 3.3

(a): On the one hand, suppose that $x \in \cup\omega$. Then x is an element of a natural number. Thus either $x \in n$ or $x = n$. Now we prove the following statement: every element of every natural number is a natural number. The base case vacuously holds, as $\emptyset$ has no elements. Now suppose the statement holds for some n. Then $n^+ = n \cup \{n\}$. Hence, if $y \in n^+$, either $y \in n$ or $y = n$. In the former case, by the induction assumption, it follows that y is a natural number, and in the latter case, y would be a natural number (as n is a natural number). Thus, in any case, y is a natural number, therefore completing the proof. As a consequence of having proven this statement, it follows that x is a natural number, and hence $x \in \omega$. On the other hand, suppose that $x \in \omega$. Thus, for some n, $x = n$. Now, all elements of n^+ are in $\cup\omega$. But n is an element of n^+. Thus $n \in \cup\omega$, and hence $x \in \cup\omega$. Thus we have proven that $\cup\omega = \omega$.

(b): On the one hand, suppose that $x \in \cup\omega^+$. Now, $\omega^+ = \omega \cup \{\omega\}$. Thus either x is an element of a natural number or x is a natural number. By the results of part (a) of this problem, it follows that if x is an element of a natural number, then x is a natural number. Thus x is a natural number, that is, $x \in \omega$. Now suppose that $x \in \omega$. Then $x \in \cup\omega^+$, as x would be an element of ω, which is an element of ω^+.